

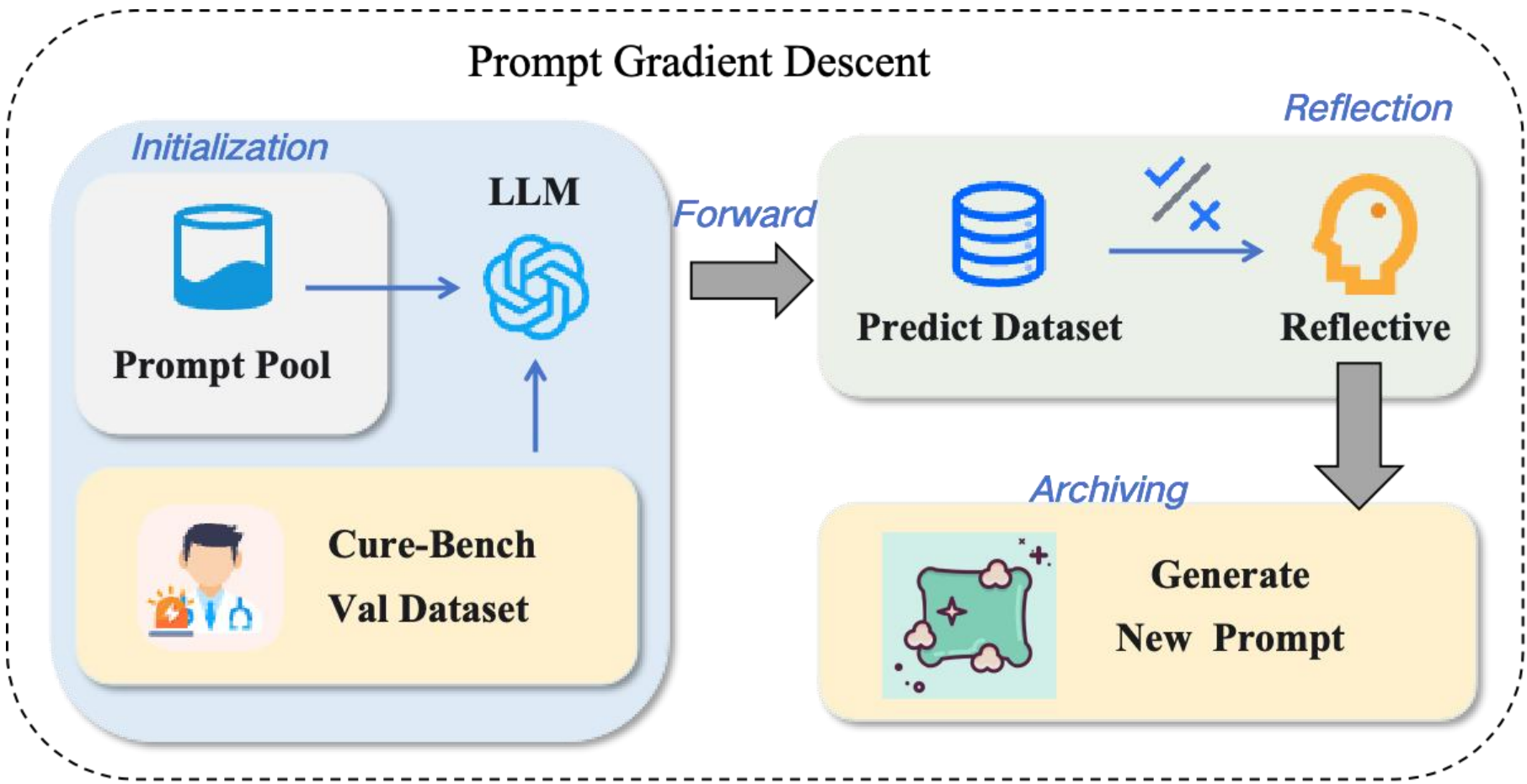
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Introduction

CURE-Bench, a competition and benchmark for evaluating AI models in drug decision-making and treatment planning. CURE-Bench includes clinically grounded tasks such as recommending treatments, assessing drug safety and efficacy, designing treatment plans, and identifying repurposing opportunities for diseases with limited therapeutic options. If only the prompt call large model for medical QA is constructed, the effect is often limited. In the era of large-scale pretraining, the base models are trained on a large amount of data, By fine-tuning downstream tasks, the model can achieve stronger performance in this task. In the current era of large language models, we can freeze the large language models and design prompts to enable them to do what we want. Similarly, we can use prompt gradient descent to make prompts and large language models more suitable for the tasks of CURE-Bench.

Main Method:

- **Initialization:** The algorithm starts with a "Candidate Pool" that initially has a single system prompt
- **Forward:** The data set results are calculated through the llm optimization module for forward propagation.
- **Reflection:** Run this new candidate prompt word on the training sample, an existing candidate prompt, and its execution result on task to generate a new variant prompt.
- **Archiving:** If a new variant prompt performs better than its parent in small batches, it is considered a successful "evolution".



Data Generation

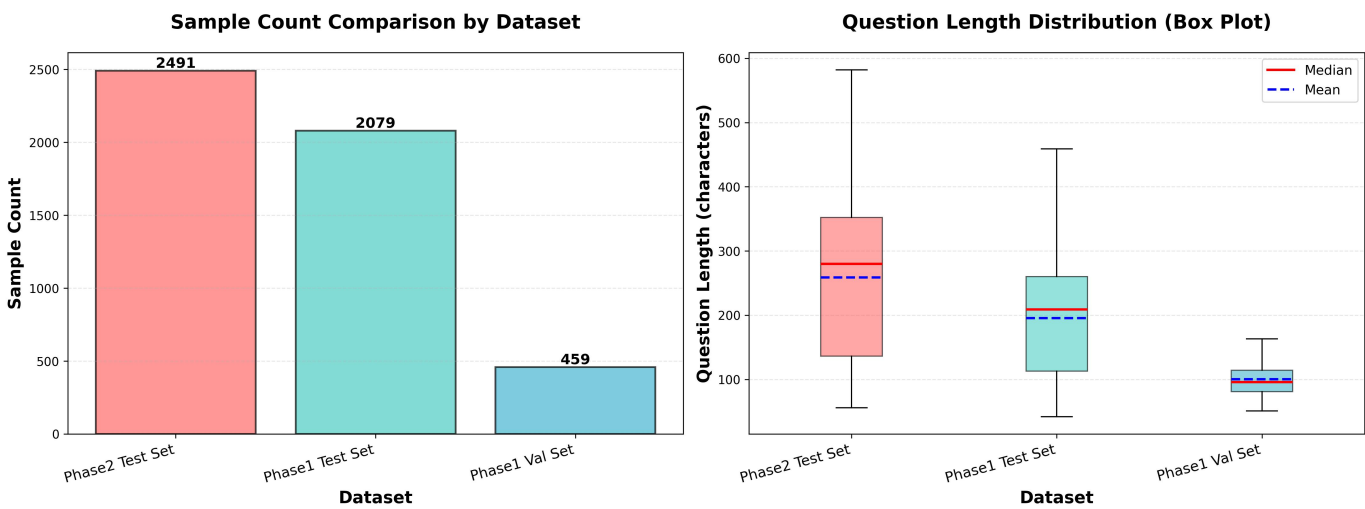
1. The dataset sample size increases across phases:

- Phase1 validation set: 459 samples
- Phase1 test set: 2,079 samples
- Phase2 test set: 2,491 samples
- Total: 5,029 samples

2. Question length distribution:

- Phase2 test set: mostly 301 – 400 characters (32.5%)
- Phase1 test set: mostly 201 – 300 characters (44.7%)
- Phase1 validation set: mostly 0 – 100 characters (58.0%)

The Phase1 validation set contains the shortest questions, and its length distribution differs from the test sets. From Phase1 to Phase2, the average question length increased by 63.3 characters, a growth of 32.4%.



Experimental Results

The dataset curebench_valset_pharse1.json was used for prompt optimization:

- **Randomly selected:** We randomly selected 100 samples from the cure-bench val dataset as the training samples for prompt optimization 0.72950
- **Hard samples:** We trained using the cure-bench val dataset through gpt5, taking 71 samples with incorrect predictions as the training samples for prompt optimization 0.73950
- **Few-shot leaning:** Based on the prompt trained by Hard samples, we added the results that were still inaccurate in the validation set as the fewshot construction prompt 0.74399

